T.I.P. TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES
COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	07/29/26	**DATE SUBMITTED:**	07/29/26
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SEATWORK NO. 3.1 Array-Based vs. STL Stack Implementations

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Quillbot, Google Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction and summarization**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Since I'm not so good at constructing words, I tend to rely on grammar checkers to easily explain my answers.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Objective
Implement a Stack Data Structure using two distinct approaches in C++: a custom Static Array implementation and the C++ Standard Template Library

Tasks & Intended Learning Outcomes (ILOs):
ILO A: Implement a stack using the C++ STL (std::stack).

ILO B.1: Implement a stack using a Static Array.

Submission Requirements:
Documentation: Please use the official Activity Template for your write-up.

Source Code: Submit your C++ file named as LastName_Seatwork4_1.cpp.

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ILO A:

Source Code:

ILOA > hare_seatwork4p1.cpp > main()

```cpp
int main(){

    // push
    // pop
    //isEmpty
    //isFull
    //Top

    //create an object
    std::stack<char> stack1;

    std::cout<<"Is the stack empty? " <<stack1.empty() <<std::endl;
    stack1.push('D');
    std::cout<<"Current top of stack: " <<stack1.top() <<std::endl;
    stack1.push('R');
    std::cout<<"Current top of stack: " <<stack1.top()<<std::endl;
    stack1.push('E');
    std::cout<<"Current top of stack: " <<stack1.top()<<std::endl;
    stack1.push('I');
    std::cout<<"Current top of stack: " <<stack1.top()<<std::endl;
    std::cout<<"Is the stack empty? " <<stack1.empty()<<std::endl;
    std::cout<<"What's the size of the stack: " <<stack1.size()<<std::endl;

    std::stack<char> stack2;

    std::cout<<"Is the stack empty? " <<stack2.empty() <<std::endl;
    stack2.push('D');
    std::cout<<"Current top of stack: " <<stack2.top() <<std::endl;
    stack2.push('A');
    std::cout<<"Current top of stack: " <<stack2.top()<<std::endl;
    stack2.push('D');
    std::cout<<"Current top of stack: " <<stack2.top()<<std::endl;
    std::cout<<"Is the stack empty? " <<stack2.empty()<<std::endl;
    std::cout<<"What's the size of the stack: " <<stack2.size()<<std::endl;

```

```
    // swap
    std::cout << "\nBefore swap:" << std::endl;
    std::cout << "Stack1 top: " << stack1.top() << std::endl;
    std::cout << "Stack2 top: " << stack2.top() << std::endl;

    stack1.swap(stack2);

    std::cout << "After swap:" << std::endl;
    std::cout << "Stack1 top: " << stack1.top() << std::endl;
    std::cout << "Stack2 top: " << stack2.top() << std::endl;

    // emplace
    stack1.emplace('X');

    std::cout << "After emplace('X') into Stack1:" << std::endl;
    std::cout << "Stack1 top: " << stack1.top() << std::endl;
    std::cout << "Stack1 size: " << stack1.size() << std::endl;

    return 0;
}
```

Output:

```
Current top of stack: E
Current top of stack: I
Is the stack empty? 0
What's the size of the stack: 4
Is the stack empty? 1
Current top of stack: D
Current top of stack: A
Current top of stack: D
Is the stack empty? 0
What's the size of the stack: 3

Before swap:
Stack1 top: I
Stack2 top: D
After swap:
Stack1 top: D
Stack2 top: I
After emplace('X') into Stack1:
Stack1 top: X
Stack1 size: 4
```

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ILO B.1: Implement a stack using a Static Array.

ILOB1 > hare_seatwork4p1p2.cpp > main()

```cpp
#include <iostream>

// Global declaration
#define maxSize 10

int stackArr[maxSize];
int top = -1;
int i, newData;

// Function prototypes

// Observer operators
bool isEmpty();
bool isFull();
void Top();
void displayAll();

// Transformer operators
void push();
void pop();

int main() {
    int choice;

    while (true) {
        std::cout << "========================";
        std::cout << "      STACK MENU\n";
        std::cout << "========================";
        std::cout << "1. Push";
        std::cout << "2. Pop";
        std::cout << "3. Is Empty";
        std::cout << "4. Is Full";
        std::cout << "5. Top";
        std::cout << "6. Display All";
        std::cout << "7. Exit";
        std::cout << "Enter your choice: ";
        std::cin >> choice;
```

```
39          switch (choice) {
40              case 1:
41                  push();
42                  break;
43              case 2:
44                  pop();
45                  break;
46              case 3:
47                  std::cout << "Is the stack empty? "
48                            << (isEmpty() ? "Yes" : "No") << std::endl;
49                  break;
50              case 4:
51                  std::cout << "Is the stack full? "
52                            << (isFull() ? "Yes" : "No") << std::endl;
53                  break;
54              case 5:
55                  Top();
56                  break;
57              case 6:
58                  displayAll();
59                  break;
60              case 7:
61                  std::cout << "Exiting program..." << std::endl;
62                  return 0;
63              default:
64                  std::cout << "Invalid choice!" << std::endl;
65          }
66      }
67
68      return 0;
69  }
```

```cpp
// Observer operators
bool isEmpty() {
    return (top == -1);
}
bool isFull() {
    return (top == maxSize - 1);
}

void Top() {
    if (isEmpty()) {
        std::cout << "Stack is empty!" << std::endl;
        return;
    }

    std::cout << "Top element: " << stackArr[top] << std::endl;
}

void displayAll() {
    if (isEmpty()) {
        std::cout << "Stack is empty!" << std::endl;
        return;
    }

    std::cout << "Stack contents (Top to Bottom): ";

    for (i = top; i >= 0; i--) {
        std::cout << stackArr[i] << " ";
    }

    std::cout << std::endl;
}

// Transformer operators

void push() {
    if (isFull()) {
        std::cout << "Stack Overflow! Cannot push another element."
                  << std::endl;
        return;
    }
```

```
112     std::cout << "Enter a value to push: ";
113     std::cin >> newData;
114 
115     stackArr[++top] = newData;
116 
117     std::cout << newData << " pushed successfully." << std::endl;
118 }
119 
120 void pop() {
121     if (isEmpty()) {
122         std::cout << "Stack Underflow! Stack is empty."
123                   << std::endl;
124         return;
125     }
126 
127     std::cout << "Popped element: " << stackArr[top] << std::endl;
128     top--;
129 }
```

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